

Caterina Morelli

✉ c.morelli@univda.it

in LinkedIn

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Profile

Research Fellow at the University of Valle d'Aosta, I recently earned my Ph.D. in Economics, Statistics and Data Science at University of Milan-Bicocca. My research combines spatial and spatiotemporal methods to study territorial heterogeneity and dynamic in energy use, GHG emissions, corporate sustainability, and tourism flows. I leverage advanced statistical and econometric tools, including spatial and spatiotemporal panel models, difference-in-differences frameworks, event-study analysis, GLMs/GAMs, spline-based spatiotemporal interactions, hierarchical clustering, and gravity models.

Education

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|---|----------------------|
| Ph.D. Economics, Statistics and Data Science
University of Milano-Bicocca, Milan, Italy
Supervisors: Prof. Paolo Maranzano, Prof. Philipp Otto (University of Glasgow)
Thesis: "Sustainability by Proximity: How Geographic Closeness Drives Corporate and Regional Carbon Emissions" (available online 🔗)
PhD with honours | Nov 2021 – Feb 2026 |
| M.Sc. Statistical, Financial and Actuarial Sciences
Alma Mater Studiorum – University of Bologna, Rimini, Italy
Supervisor: Prof. Enrico Supino
Thesis: "The impact of sustainability awards on firm's market value: the case of SEAL Business Award"
110/110 with honours | Nov 2018 – Oct 2020 |
| B.A. Economics and Commerce
Marche Polytechnic University, Ancona, Italy
Supervisor: Prof. Francesco Maria Chelli
Thesis: "Wellbeing is not evaluated in euros: correlation between Pil and Bes"
110/110 with honours | Sep 2015 – July 2018 |

Visiting Periods

University of Valencia, ES	April - May 2026
University of Glasgow, UK	April - Sep 2024
Bocconi University, Milan, Italy	Oct 2023 – Jan 2024
Erasmus+: Universitat Jaume I, Castellon de la Plana, Spain	Sept 2017 - Jan 2018

Other educational experience

Postgraduate Course: Network Econometrics, SIde, Ca' Foscari University of Venice	July 2025
Summer School: Generalized Additive Models in R, Ca' Foscari University of Venice (with Simon N. Wood, University of Edinburgh)	June 2025
Interdisciplinary course: Introduction to grant writing, UniMiB	Apr 2025
PhD Course: Kalman Filter and State Space Models, University of Padua (with Siem Jan Koopman, VU University Amsterdam)	June 2024
Interdisciplinary course: Literature review: managing and evaluating your bibliography, UniMiB	Apr 2024
Summer School: Macroeconomic Forecasting and Analysis in the Machine Learning Era, SIde (Perugia, Italy)	July 2023

PhD Course: Multivariate analysis, Bocconi University	Oct 2023
PhD Course: Machine Learning, Bocconi University	Oct 2023
PhD Course: Time Series Analysis using UCM and State Space Models, University of Aveiro, Portugal (online)	May 2023

Work Experience

Research fellow - University of Valle d'Aosta	Dec 2025 - ongoing
Project SMARTDECO: Energy and Tourism in Fragile Mountain Areas	
Responsible Prof. Consuelo Rubina Nava	

Research Assistant

University of Milano-Bicocca	
WebSite Premia: Empirical analysis of Italian firms' performance	Dec 2024 - June 2025
Responsibles: Prof Piergiiovanna Natale, Prof Valeria Gattai	
CIPN Care Project: Analysis of hospital data in Stata	Feb – July 2023
Responsible: Prof Paola Alberti	

Teaching Assistant

Bocconi University	
Statistics – Bachelor's degree – Prof. Raffaella Piccarreta	2024/25
University of Milano-Bicocca	
Econometrics (in English) – Master's degree – Prof. Luca Stanca	2024/25
Statistics II – Bachelor's degree – Prof. Riccardo Borgoni	2024/25
Statistics II – Bachelor's degree – Prof. Riccardo Borgoni	2023/24
Statistics II – Bachelor's degree – Tutoring for student with SLD	2022/23
University of Milan (Statale)	
Statistics – Bachelor's degree – Prof. Daniele Spinelli	2024/25
Statistics – Bachelor's degree – Prof. Andrea Ariu	2023/24
Statistics (in English) – Bachelor's degree – Prof. Silvia Salini	2022/23
Mathematics – Bachelor's degree – Prof. Giovanni Puccetti	2022/23
Statistics – Bachelor's degree – Prof. Chiara Tommasi	2022/23

Other Professional Experience

Analyst, Deloitte Consulting S.r.l.	Dec 2020 – Oct 2021
Monitoring and improvement of IT systems for investment and insurance products	
Volunteer, Giramundo Project (AIESEC)	July – August 2019
Centro Educacional Dom Bosco, Natal, Brazil	

Conferences and Workshops

ERSA , Sofia University, Bulgaria*	Aug 2026
EcoMod , University of Luxembourg, Luxembourg*	August 2026
SIS-FEN 2026 , Sapienza University of Rome, Italy*	June 2026
Milan PhD Economic Workshop , Bocconi University, Italy*	May 2026
Seminar , University of Valencia, Spain*	May 2026
ICES SIS , University of Bari, Italy*	Feb 2026
IWEEE Side , Marche Polytechnic University, Italy [◊]	Jan 2026
AIRe , University of Chieti-Pescara, Italy*	Sept 2025
IES , University of Padua, Bressanone-Brixen* [†]	June 2025
ICEEP (CefES) , University of Milano-Bicocca, Italy*	June 2025

SEEDS, Catholic University of the Sacred Heart, Milan, Italy*	May 2025
ReLunch Seminar, University of Milano-Bicocca, Italy*	May 2025
SEA, Mohammed V University in Rabat, Morocco ^{§*}	May 2025
IAERE, Roma Tre University, Rome, Italy*	Feb 2025
CfeCms, King's College London, UK [‡]	Dec 2024
Milan PhD Economic Workshop, University of Milano-Bicocca, Italy*	Sept 2024
MBC2, University of Catania, Italy [◇]	Aug 2024
METMA XI, Lancaster University, UK [◇]	July 2024
ICEEP (CefES), ETH Zürich, Switzerland*	June 2024
Statistics Seminar, University of Glasgow, UK*	June 2024
PhD Seminar, University of Milano-Bicocca, Italy*	Feb 2024
IWEEE Side, Free University of Bozen-Bolzano, Italy*	Jan 2024
ERSA Congress, University of Alicante, Spain*	Aug 2023
Milan PhD Economic Workshop, University of Milan (Statale), Italy*	June 2023

*Presenter, [§]Discussant, [‡]Invited session, [‡]Session organizer, [◇]Poster.

Research Projects

- SMARTDECO – Energia e Turismo in Aree Montane Fragili: Strategie e metodologie per il Decoupling** 2025 – 2027
 Research project funded under the “Aggregazioni R&S – Transizione ecologica” call (FESR Valle d'Aosta 2021/2027). Università della Valle d'Aosta. Scientific coordinator: Prof.ssa Consuelo Rubina Nava.
- Website Premia** 2024
 Research collaboration on the evaluation of Italian firms' website quality and its relationship with firms' performance. Department of Economics, Quantitative Methods and Business Strategies, University of Milano-Bicocca. Scientific coordinators: Prof. Piergiorgio Natale, Prof. Valeria Gattai.
- CIPN CARE – Peripheral neuropathy from chemotherapy: why should we care?** 2022 – 2023
 Collaboration within the research project of the Department of Medicine and Surgery, University of Milano-Bicocca. Scientific coordinator: Dott.ssa Paola Alberti.

Working Papers

- C. Morelli, P. Maranzano, R. Mattera P. Otto. *Cluster-wise spatiotemporal panel models for global and local spillovers identification*. Preliminary draft available upon request
- M. Longhi, C. Morelli (2025). *The role of Chinese Panda Ambassadors in International Trade*. DEMS Working Paper: 550 [🔗](#)

Peer-reviewed Articles

1. P. Maranzano, R. Mattera, C. Morelli, G. Scepti (2026). Forecast reconciliation of agricultural GHG emissions in Europe with fuzzy clustering. *Annals of Operations Research* doi.org/10.1007/s10479-026-07363-y [🔗](#)
2. C. Morelli, P. Maranzano, P. Otto (2026). Spatiotemporal clustering of GHGs emissions in Europe: exploring the role of spatial component. *Spatial Statistics*. doi.org/10.1016/j.spasta.2026.100960 [🔗](#)
3. S. Boccaletti, P. Maranzano, C. Morelli, E. Ossola (2025). ESG Performance and Stock Market Responses to Geopolitical Turmoil: evidence from the Russia-Ukraine War. *Economic Modelling* doi.org/10.1016/j.econmod.2025.107380 [🔗](#)
4. C. Morelli, S. Boccaletti, P. Maranzano, P. Otto (2025). Multidimensional spatiotemporal clustering – An application to environmental sustainability scores in Europe. *Environmetrics*. doi.org/10.1002/env.2893 [🔗](#)
5. R. Borgoni, M. Denova, P. Maranzano, C. Morelli (2023). Assessing geographical differences of the COVID-19 impact on fuel consumption: the case of Italy. *Letters in Spatial and Resource Sciences*. 10.1007/s12076-023-00361-9 [🔗](#)

Short paper and Book of abastracs

1. C. Morelli (2026). Tourist Spending Capacity and Greenhouse Gas Emissions: Evidence from European NUTS2 Regions. *Book of short papers*. ICES 2026 - 4th International Conference on Economic Statistics [🔗](#)

Reviewer for peer-reviewed journals

Economic Modelling
Annals of Operations Research
International Journal of Climate Change Strategies and Management
Humanities and Social Sciences Communications
Economics of Disaster and Climate Change
Communications in Statistics - Theory and Methods.

Scholarships

Erasmus+ scholarship for research period abroad (2024)
Ph.D. 4-year scholarship, University of Milano-Bicocca (2021–2025)
Erasmus+ scholarship for study period abroad (2017)

Languages

Italian (native)
English (fluent)
Spanish (fluent)
French (intermediate)

Software & Programming

Advanced: Microsoft Office / 365, R, $\text{\LaTeX}$
Intermediate: VBA, Stata, Python, html
Basic: MySQL, Gretl, Matlab, Dynare

SCDA R package contribution: HSTC functions for hierarchical spatiotemporal cluster analysis of high-dimensional data; available in the upcoming release SCDA package documentation (CRAN) [↗](#).